Generalized Morse theory for distance functions

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1 Generalities on Lipschitz Functions

1.1 Lipschitz functions on manifolds and subdifferentials

For all the following, M will denote a differentiable manifold of dimension m. In this section, we generalize many definitions related to Lipschitz functions. We begin by extending the definition of Lipschitz functions to functions on a manifold.

Definition 1. *A function $f : M \to \mathbb{R}^n$, where M is a smooth manifold, is locally Lipschitz if for any chart $U \subset M$ and any compact $K \subset U$, there exists $C > 0$ such that, with the coordinates on U*

$$\forall x, y \in K, |f(x) - f(y)| \leq C|x - y|$$

Let $f : U \subset \mathbb{R}^m \to \mathbb{R}^n$ be a Lipschitz function, where U is open. Rademacher's theorem, which states that f is differentiable almost everywhere, will allow us to generalize differentials to this setting.

Definition 2. *[3] The subdifferential of f at x, denoted by $\partial_x f$ is the convex hull of the linear maps l obtained as*

$$l = \lim_{i \to \infty} d_{x_i} f$$

where f is differentiable at each x_i and $x_i \to x$.

The subdifferential is then set valued. This set is always non empty and compact, f being Lipschitz bounds df where it is defined.
By [2], we can exclude a set of measure 0 from the possible values of x_i without changing the definition.

Proposition 1. *∂f is the smallest convex valued upper semi-continuous function containing df whenever it is defined.*

Proof. First, ∂f is upper semi-continuous : by compactness, is suffices to show that for all $x \in U, \varepsilon > 0$, for y sufficiently close to x.

$$\partial_y f \subset \partial_x f + B(0, \varepsilon)$$

Assume for contradiction that there exists a sequence $x_i \to x$ such that for all i, $\partial_{x_i} f$ has an element out of $A := \partial_x f + \overline{B}(0, \varepsilon/2)$. A being convex and closed, let x_i' be such that $d_{x_i'} f$ is out of A, and such that $x_i' - x_i \to 0$. Up to extraction, $d_{x_i'} f$ converges to an element in $\partial_x f$, so it enters A, contradiction.
The second part follow from ∂f being a convex hull. □

The subdifferential allows us to generalize the idea of regular and critical points, which will be an important tool for generalizing Morse theory. The definition is as follows :

Definition 3. *A point $x \in U$ will be called critical if there exists some $l \in \partial_x f$ not of maximal rank. x will be called regular if it isn't critical.*

To extend these subdifferentiability notions to manifolds, one must guarantee that for all charts, the definitions of ∂f coincide. In this is the case, $\partial_x f$ is a compact convex subset of $T_x^* M$. The following generalization of the chain rule allows us to do that.

Proposition 2 (Subdifferential chain rule)**.** *Let $h : \mathbb{R}^n \to \mathbb{R}^k$ be a C^1 function. For $x \in U$:*

$$\partial_x (h \circ f) \subset d_{f(x)} h \circ \partial_x f$$

If h is a diffeomorpism, we have equality.

Proof. For the first part, because f is differentiable a.e, it suffices to show that the leftwise term is an upper semi-continuous convex valued function that contains $d(h \circ f)$ whenever df is defined. This follows from the definition of ∂f.

If h is a diffeomorphism, then for $x \in U$:

$$\partial_x f = \partial_x (h^{-1} \circ h \circ f) \subset d_{h \circ f(x)} h^{-1} \circ \partial_x (h \circ f) \subset \partial_x f$$

so we have equalities, and the result follows. □

Although some important differentiablity related results can be obtained with locally Lipschitz functions as we'll see, a general locally Lipschitz function can have some pathological behaviours. As an example, it's subdifferential isn't in general a topological submanifold of M as illustrated by Figure ??. However, the functions we'll study, for example continuous selections (see Section 2) have much nicer behaviour and we retrieve this property.

1.2 Generalized differential theorems

We now consider the specific case of $f : U \subset \mathbb{R}^m \to \mathbb{R}^m$. This section is devoted to proving generalizations by Clarke [3] of the inverse function theorem and the implicit function theorem for the case of Lipschitz functions.

In this section, we norm the space of linear maps $\mathbb{R}^m \to \mathbb{R}^m$, which we denote $\mathcal{L}(\mathbb{R}^n)$, with the operator norm $\|\cdot\|$ obtained from the norm used on $\mathbb{R}^n$.

Lemma 1. *Assume U is convex, and that there exists some closed convex set C such that*

- *Every $l \in C$ is invertible.*
- *$\partial_x f \subset C$ for all $x \in U$*

Then there exists some $\delta > 0$ such that for all $x, y \in U$:

$$|f(x) - f(y)| \geq \delta |x - y|$$

Proof. Firstly, we set

$$\delta = \inf \{|l(h)| \mid l \in C, h \in S\}$$

where S is the unit sphere in $\mathbb{R}^n$. $\delta > 0$ because all $l \in C$ are invertible and $C \times S$ is closed.

Let $x, y \in U$ and assume $x \neq y$. Let π be the plane perpendicular to $y - x$ passing through x. f being differentiable a.e, Fubini's theorem implies that for almost all $x' \in \pi$, f is differentiable almost on the whole line passing through x' and perpendicular to π (i.e f isn't differentiable on a subset of this line of one dimensional measure 0).
Denoting $F : t \longmapsto f(x' + t(y - x))$, we have :

$$f(x' + y - x) - f(x') = \int_0^1 F'(t)dt = \int_0^1 d_{x'+t(y-x)} f(y - x)dt \tag{1}$$

Define the linear map

$$L = \int_0^1 d_{x'+t(y-x)} f dt$$

C being convex and closed, $L \in C$. Observe that the integral on the right of (1) is just $L(y - x)$. So

$$|f(x' + y - x) - f(x')| \geq \delta |y - x|$$

This inequality holds for almost all $x' \in \pi$, so by continuity of f, it holds for x and we get the desired result. □

Theorem 1 (Lipschitz inverse function theorem)**.** *If x_0 is a regular point of f, then there exists V, W, neighbourhoods of x_0 and $f(x_0)$ respectively, such that f induces a bijection $f : V \to W$ where f^{-1} is Lipschitz too.*

Proof. By compactness of $\partial_{x_0} f$, regularity of x_0, and the fact that the invertible linear maps form an open set, there exist some $\varepsilon > 0$ such that the set

$$C := \partial_{x_0} f + \overline{B}(0, \varepsilon)$$

is composed only of invertible maps. By upper semi-continuity of ∂f, there is some $r > 0$ such that for $y \in B(x_0, r)$, $\partial f(y) \subset C$. We then apply Lemma 1 with $U = B(x_0, r)$ and get a corresponding $\delta > 0$.

Set $V = B(x_0, r/2)$ and $W = f(V)$. We showed in particular that f is injective on $\overline{V}$, by compactness $f : \overline{V} \to f(\overline{V})$ induces a homeomorphism. W is then open. The inequality we got with δ implies that f^{-1} is $\frac{1}{\delta}$ Lipschitz. $\square$

From this theorem, follows the implicit function theorem for Lipschitz functions

Theorem 2. *Let $F : U \to \mathbb{R}^n$, where $U \subset \mathbb{R}^m \times \mathbb{R}^n$ is open and F is Lipschitz. Let $(x_0, y_0) \in U$ such that $F(x_0, y_0) = 0$ and y_0 is a regular point of $y \mapsto F(x_0, y)$. Then there exist neighbourhoods $(x_0, y_0) \in W \subset U$, $x_0 \in V \subset R^m$ and a Lipschitz function $g : V \to \mathbb{R}^n$ such that*

$$(x, y) \in W \text{ and } F(x, y) = 0 \Leftrightarrow x \in V \text{ and } y = g(x)$$

The proof is almost the same as the usual implicit function theorem, where the continuity of the differential is replaced by the semi continuity of ∂F.

1.3 First Morse lemma for locally Lipschitz functions

The goal of this section is to use the generalized notion of critical points in order to generalize the first Morse lemma to the case of locally Lipschitz functions. Fix M an m-dimensional smooth manifold, $f : M \to \mathbb{R}$ a continuous function and define for all $c \in \mathbb{R}$ the sublevel set of f of value c as

$$M_c := f^{-1}(]-\infty, c])$$

The classical first Morse lemma states that if f is C^1, and there are no critical points in $f^{-1}([a, b])$ (that we assume to be compact), then M_b is diffeomorphic to M_a and M_a is a deformation retract of M_b. We extend this to locally Lipschitz functions, following [1]. Note that in this setting x is a critical point of f iff $0 \in \partial_x f \subset T_x^* M$.

Proposition 3. *Let $a < b \in \mathbb{R}$ such that $f^{-1}([a, b])$ is compact and free of critical points. Then there exists a locally Lipschitz homotopy $H : M_b \times [0, 1] \to M_b$ such that for all $s \in [0, 1]$, $H(\cdot, s)$ is a homeomorphism verifying*

$$H(M_b \times \{s\}) \subset M_{b(1-s)+as}$$

Proof. We'll construct $V \in \mathrm{Vec}(M)$ a smooth complete vector field on M such that, for all $x \in M$ such that $d_x f$ is defined :

$$d_x f(V(x)) \leq 0 \text{ and } d_x f(V(x)) \leq -1 \text{ if } x \in f^{-1}([a, b]) \tag{2}$$

Let $x \in f^{-1}([a, b])$. x being regular and $\partial_x f$ convex, we find an affine hyperplane in $T_x^* M$ separating it from 0. This gives us $v_x \in T_x M$ such that $\langle \partial_x f, v_x \rangle \leq -2$. By semi continuity, there exists an open neighbourhood U_x of x such that for all $y \in U_x$, $\langle \partial_y f, v_x \rangle \leq -1$. We choose this neighbourhood such that it has compact closure.

These U_x cover $f^{-1}([a, b])$. Let $U_{x_1}, ..., U_{x_r}$ be a finite subcover, and let $\rho_1, ..., \rho_r$ be a partition of unity subordinated to this covering. We set

$$\forall x \in M, V(x) = \sum_{i=1}^{r} \rho_i(x) v_{x_i}$$

which indeed verifies (2) : if $d_x f$ is defined, it is contained in $\partial_x f$, and V is of compact support so it is complete.

Now observe that for $x \in f^{-1}([a, b])$, $t \mapsto f(e^{tV} x)$ is strictly decreasing. Denote by $t(x, s)$, $s \in [0, 1]$, the unique time such that $f(e^{t(x,s)V} x) = as + f(x)(1 - s)$. The equation defining $t(x, s)$ is locally Lipschitz in s, x and t, so by the Lipschitz implicit function theorem, t is a locally Lipschitz function of (x, s).

We can similarly define t' a locally Lipschitz function of x such that $f(e^{t'(x)V}x) = b$ for x in the orbit of some $y \in f^{-1}(b)$ by V. Denote O the set of $x \in M$ that are in such orbits. The homotopy is then given by

$$H : \begin{array}{ccc} M_b \times [0,1] & \longrightarrow & M_b \\ (x,s) & \longmapsto & \begin{cases} e^{t(e^{t'(x)V}x,s)V}x & \text{if } x \in O \\ x & \text{else} \end{cases} \end{array}$$

□

2 Morse theory of continuous selections

2.1 Definitions

We now restrict ourselves to continuous functions that are piecewise smooth with finitely many pieces. Following (ref), we define those as continuous selections.

Definition 4. *A continuous function $f : M \to \mathbb{R}$ is a continuous selection if there exists $\varphi_1, ..., \varphi_n$ "regular" functions such that*

$$\forall x \in M, f(x) \in \{\varphi_1(x), ..., \varphi_n(x)\}$$

The term "regular" is used loosely here to indicate that the φ_i satisfy some extra condition on top of being continuous. For what follows, we'll restrict ourselves to continuous selections of smooth functions (but C^2 will suffice).

Note that a continous selection of locally Lipschitz functions is itself locally Lipschitz : if M is an open set of $\mathbb{R}^m$, and we assume that the functions φ_i are Lipschitz with constants c_i, then f is Lipschitz with constant $\max_i c_i$. Because smooth functions are locally Lipschitz, the functions we'll study will be too.

We can then relate ∂f to the differentials of the φ_i. Indeed, a limit of points of the form $d_x\varphi_i$ in T^*M is of the same form. Define the set of active indices at x,

$$I(x) = \{i \in [\![1, n]\!] \mid \varphi_i(x) = f(x)\}$$

$\partial_x f$ is then the convex hull of some $d_x\varphi_i$, $i \in I(x)$. It is possible that not all $d_x\varphi_i$ are used, if for example one of the functions isn't active in a neighbourhood of x.

We would like to define what it means for such a continuous selection to be Morse. Let f be a continuous selection of $\varphi_1, ..., \varphi_n$. Following [1], we give three conditions for a function to be considered Morse.

$$\text{For all } i \neq j, \text{ the graphs of } \varphi_i \text{ and } \varphi_j \text{ are transversal.} \tag{C1}$$

This condition has some important consequences : for a non empty $I \subset [\![1, n]\!]$, we define $W_I := \{x \in M \mid I(x) = I\}$. Condition (C1) implies that W_I is a submanifold of M of codimension $|I| - 1$: the intersection of the graphs of $\varphi_i, i \in I$ is a submanifold of $M \times \mathbb{R}$ of codimension $|I|$ and W_I is the pre-image of this intersection through the diffeomorphism $x \mapsto (x, \varphi_i(x))$, $i \in I$, with the codomain being the graph of φ_i.

This transversality condition also implies that for $x \in M$, the differentials $d_x\varphi_i$ are affinely independant where $i \in I(x)$. In particular, $\partial_x f$ is a simplex with less than $|I(x)| - 1$ vertices.

Note that for all $I \subset [\![1, n]\!]$, $f_{|W_I}$ is a smooth function. This motivates the second condition

$$\text{For all non empty } I \subset [\![1, n]\!], f_{|W_I} \text{ is Morse} \tag{C2}$$

Consider $x \in M$ a critical point of f. Observe that x must also be a critical point of $f_{|W_{I(x)}}$. So $T_xW_{I(x)} \subset \bigcap_{i \in I(x)} d_x\varphi_i$, but the converse can fail if the $d_x\varphi_i$ are not linearly independant enough. The last condition fixes that

$$\forall x \in M, \forall j \in I(x), (d_x\varphi_i)_{\substack{i \in I(x) \\ i \neq j}} \text{ are linearly independant} \tag{C3}$$

Geometrically, this condition corresponds to requiring that 0 is either outside or in the interior of the simplex $\mathrm{conv}\{d_x\varphi_i \mid i \in I(x)\}$, but not on one of its faces. Note (i think ??) that this condition can be relaxed by restricting it to the subset of $I(x)$ participating in $\partial_x f$.

A continuous selection f is then called a Morse function if it verifies all three conditions. Note that as in the smooth case, a generic continuous selection is Morse : (C1) is a generic condition, any two functions can be slightly perturbed in order for their graphs to be transversal, and any small perturbation of transversal graphs preserves transversality. By genericity of smooth Morse functions, (C2) is also a generic condition. With the geometric interpretation, we can also see that condition (C3) is generic.

2.2 Normal form of a Morse continuous selection

We show here an equivalent of the second Morse lemma for the case of Morse continuous selections. Noticing that $x \mapsto I(x)$ is upper semi-continuous and using the definitions yields the following result.

Lemma 2. *If $f : M \to \mathbb{R}$ is a Morse continuous selection, all its critical points are isolated.*

We'll then consider, up to restricting ourselves to a small enough chart, that $f : \mathbb{R}^m \to \mathbb{R}$ is Morse and has no critical points other than 0. We'll write for shorthand $W := W_{I(0)}$, $k := |I(0)| - 1$, $c := f(0)$, and assume that $I(0) = [\![1, k+1]\!]$.

Theorem 3. *If 0 is a critical point of f, then there exists Lipschitz local coordinates around 0, $\psi : \mathbb{R}^k \times \mathbb{R}^{m-k} \to U$ such that*

$$\forall y \in \mathbb{R}^k, \forall z \in \mathbb{R}^{m-k}, f(\psi(y,z)) = \lambda(y) + q(z)$$

where λ is a continuous selection of $k+1$ affinely independant linear functions, and q is of the form

$$\forall z \in \mathbb{R}^{m-k}, q(z) = c - \sum_{i=1}^{\mu} z_i^2 + \sum_{i=\mu+1}^{m-k} z_i^2$$

with $0 \leq \mu \leq m-k$.

Proof. We first restrict ourselves to O a neighbourhood of 0 such that W can be considered rectified in some smooth coordinates i.e, $W \cap O = 0 \times \mathbb{R}^{m-k}$. We can also use the smooth Morse lemma to get $f(0,z) = q(z)$. For $(y,z) \in \mathbb{R}^m$, write $f^0(y,z) = f(y,z) - f(0,z)$. We will build for all z close enough to 0 a locally invertible map $\psi_z : \mathbb{R}^k \to \mathbb{R}^k$ such that $\psi_z(0) = 0$ and for y close enough to 0, $f^0(\psi_z(y), z)$ is a continuous selection of linear maps independant of z.

For what follows we'll fix z and omit it in the notations for convenience. We find a neighbourhood U of $0 \in \mathbb{R}^k$ such that

- U is connected.
- For all $y \in U$, $I(y) \subset [\![1, k+1]\!]$.
- For all $j \in [\![1, k+1]\!]$, the set O_j of points in U such that $f(y) \neq \varphi_j(y)$ is open.
- There exists $\psi_j : U \to V_j$ a diffeomorphism such that for all $i \neq j$, $\varphi_i(\psi_j(y)) = \varphi_i(0) + d_0\varphi(y)$

We now want to combine these to take account of the possible intersections between the O_j. Our goal is to combine to ψ_j together with coefficients $\mu_j : U \backslash \{0\} \to \mathbb{R}$ and set our final diffeomorphism to be

$$\psi(y) = \sum_{i=1}^{k+1} \mu_i(y)\psi_i(y)$$

Moreover, ψ should take all active φ_i to its affine approximation.
Let $\Omega \subset U$ be a maximal set on which this is possible. Assume there exists some non zero $y \in \Omega$ verifying that all of these exist at y. Let j be such that $y \in O_j$

$$F : \begin{array}{ccc} U \times \mathbb{R}^{k+1}_{<2} & \longrightarrow & \mathbb{R}^{k+1} \\ (y, \mu_1, ..., \mu_{k+1}) & \longmapsto & \left(\varphi_i \left(\sum_{r=1}^{k+1} \mu_r \psi_r(y) - \varphi_i(0) - d_0\varphi_i(y) \right)_{i \neq j}, \sum_{i=1}^{k+1} \mu_i - 1 \right) \end{array}$$

We'll assume for convenience for what follows that the $\psi_i(y)$ are affinely independant. A modified version of the argument also works if the ψ_i are allowed to be affinely dependant but its technicality doesn't add up much to the idea of the proof.

By assumption we have $F(y, \mu_1(y), ..., \mu_{k+1}(y)) = 0$. Then by condition C3, the differential of F along μ is invertible. So by the implicit function theorem, there exists a neighbourhood of y on which μ_i $i \neq j$ can be extended. So Ω is open in $U \backslash \{0\}$.

Ω is also closed in this set : assume there is a sequence (y_r) in Ω converging to some non zero $y \in O_j$ for some j. By construction, the μ_i are bounded, so we can assume up to extraction that $\mu(y_r) \to \tilde{\mu}$. Then $F(y, (\tilde{\mu}_i)_{i \neq j}) = 0$, and we again use the implicit function theorem to build functions $\tilde{\mu}_i$ on a ball around y. Now take any point $y' \in \Omega$ in this ball. The implicit function theorem states that the μ_i are the only functions that gives 0 when fed into F and defined on a certain ball around y'. $\tilde{\mu}_i$ also satisfies these conditions, so $\tilde{\mu}_i(y') = \mu_i(y')$, so $\tilde{\mu}_i$ extends μ_i to y, $y \in \Omega$.

Because $U \backslash \{0\}$ is connected, $\Omega = U \backslash \{0\}$ because we can trivially define μ_i at a point where only one index isn't active. ψ is then a local diffeomorphism using the fact that $d_0\varphi_i \circ d_0\psi = d_0\varphi_i$ for all i.

We constructed for all z close enough to 0 to approximate $d_{(0,z)}\varphi_i$ by $d_0\varphi_i$, ψ_z. They verify the following locally Lipschitz equation : $f(\psi_z(y), z) = \lambda(y) + q(z)$ where λ is a continuous selection of $d_0\varphi_i$. So $(z, y) \mapsto \psi_z(y)$ is locally Lipschitz.

□

We now try to understand better the structure of the non smooth component that appears in the normal form theorem. It turns out that it can be encoded in a purely combinatorial way, as the following theorem shows [4].

Theorem 4. *Let $\lambda : \mathbb{R}^k \to \mathbb{R}$ be a continuous selection of $k+1$ affinely independant linear functions, $\lambda_1, ..., \lambda_{k+1}$. Then there exists a unique family $S_1, ..., S_l$ of subsets of $[\![1, k+1]\!]$ verifying that*

- *$S_i \not\subset S_j$, $i \neq j$*
- *λ can be written as*

$$\lambda(y) = \min_i \max_{j \in S_i} \lambda_j(y)$$

Proof. Let's first tackle uniqueness. Let $S_1, ..., S_l$, $D_1, ..., D_r$ be two such distinct families. Assume without loss of generality that $S_1 \neq D_j$. We can also assume that for all $1 \leq j \leq r$, $S_1 \not\supset D_j$. If it is the case, then for all $1 < i \leq l$ we have $D_j \not\supset S_i$ by assumption on S. We can then swap S and D to get what we want.
By affine independance, the following linear system has a solution

$$\lambda_i(y) + \mu = c_i, i \in [\![1, k+1]\!]$$

where the unknowns are y, μ and $c_i = 0$ if $i \in S_1$ and 1 otherwise. It follows that, if we use the representation with D_i, we get $\lambda(y) = 1 - \mu$ and if we use the one with the S_i, $\lambda(y) = -\mu$.

Now for existence. For all permutation $\nu \in \mathfrak{S}_{k+1}$ define the the cone

$$K_\nu = \{y \in \mathbb{R}^k \mid \lambda_{\nu(1)}(y) \leq ... \leq \lambda_{\nu(k+1)}(y)\}$$

and the function $s : \nu \in \mathfrak{S}_{k+1} \mapsto i \in [\![1, k+1]\!]$ where i is such that $\lambda_{\nu(i)} = \lambda$ on K_ν. If we assume that for all y and ν we have

$$\lambda(y) \leq \max\{\lambda_{\nu(1)}(y), ..., \lambda_{\nu(s(\nu))}(y)\}$$

then it suffices to consider the family of set $S_\nu = \{\nu(1), ..., \nu(s(\nu))\}$ and simplify these by removing those who are included in one another.

We now give a broad idea of the proof of the claim. The result is true by definition on K_ν. For any $y \in \mathbb{R}^k$, set $y' \in K_\nu$ and consider the evolution of the order of λ_i along the line between y and y' : the value of each λ_i evolves at a constant rate. Whenever the current λ_i defining λ encounters another λ_j and λ becomes defined as λ_j afterwards, either λ_j is smaller than λ_i after the encounter, or the opposite. If we denote I the set of indices such that the functions are smaller than λ_j before the encounter, then in either case we see that λ stays smaller than the maximum of $\lambda_{i'}$ $i' \in I$. This concludes the proof because the line starts at K_ν. □

2.3 Describing the homology of sublevel sets

Our goal for this section is to describe more precisely the transition that happens as the critical point of f is passed, where f is still a Morse continuous selection defined on $\mathbb{R}^m$ with 0 as its only critical point.

The normal form of f and theorem 4 allows us to fully describe the behaviour of f with purely combinatorial information, a smooth Morse index and a more complicated set representation. We can encode the combinatorial structure that appears in its non smooth part in a way that will give us topological and homotopical informations about the sublevel sets : define the simplical complex $C(S_1^c, ..., S_l^c)$ on $k+1$ vertices whose faces are given by the complements in $[\![1, k+1]\!]$ of the sets S_i describing the non smooth part of f.

Another thing to note is that when put in normal form, f is a quasi homogeneous function, i.e there exists some real numbers $\alpha_1, ..., \alpha_m > 0$ such that for all $t > 0, x \in \mathbb{R}^m$, we have

$$f(t^{\alpha_1} x_1, ..., t^{\alpha_m} x_m) = tf(x)$$

In this case we have $f(ty, t^{1/2}z) = tf(y, z)$ if we assume that $f(0) = 0$, assumption we'll make without loss of generality. For a quasi homogenous function h, we'll denote δ_t the power-scaling map such that $h(\delta_t x) = th(x)$. The following lemmas on quasi homogeneous functions will prove useful for describing the homotopy type of sublevel sets.

Lemma 3. *Let $h : \mathbb{R}^m \to \mathbb{R}$ be a quasi homogeneous locally Lipschitz function having only 0 as a critical point. We have the following homotopy equivalence*

$$(M_1, M_{-1}) \simeq (C_{h^{-1}(-1)}, h^{-1}(-1))$$

where M_1, M_{-1} are the sublevel sets and $C_{h^{-1}(-1)}$ is the cone over $h^{-1}(-1)$.

Proof. Let $\mathbb{S}$ denote the unit sphere. Define a smooth V over $\mathbb{S}$ such that for all $x \in \mathbb{S}$, $V(x) \in \mathbb{R}^m$ and $\langle \partial_x h, V(x) \rangle \leq -1$. Extend V into a vector field on $\mathbb{R}$ such that it commutes with δ_t for all $t > 0$, i.e for $x \in \mathbb{S}$, $V(\delta_t x) = d_x \delta_t(V(x))$. It follows that $s \mapsto h(e^{sV}x)$ is strictly decreasing for all $x \neq 0$.

From this observation, we define similarly to lemma 3, for all $x \in h^{-1}(]-\infty, 0[)$, $p(x)$, the only point of the orbit of x in $h^{-1}(-1)$. We also define for all $x \in h^{-1}(-1), s \in [0, 1[$, $t(x, s)$ such that it verifies $h(e^{-t(x,s)V}x) = h(x) + s$. If we then denote by $\pi : [0, 1] \times h^{-1}(-1) \to C_{h^{-1}(-1)}$ the projection map, where the top of the cone is the point at height $s = 1$, the homotopy equivalence maps are the following

$$F : \begin{array}{ccc} M_1 & \longrightarrow & C_{h^{-1}(-1)} \\ x & \longmapsto & \begin{cases} \pi(0, p(x)) & x \in M_{-1} \\ \pi(h(x) + 1, p(x)) & x \in h^{-1}([-1, 0]) \\ \pi(1, 0) & x \in h^{-1}([0, 1]) \end{cases} \end{array}$$

$$G : \begin{array}{ll} C_{h^{-1}(-1)} & \longrightarrow M_1 \\ \pi(s,x) & \longmapsto e^{-t(x,s)}x \end{array}$$

We can see that $F \circ G = \mathrm{id}_{C_{h^{-1}(-1)}}$. $G \circ F$ is homotopic to the identity : we can use V to contract M_{-1} onto $h^{-1}(-1)$, and also to retract $h^{-1}([0,1])$ onto $h^{-1}(0)$. □

If X and Y are two topological spaces, we define the join of X and Y, denoted by $X * Y$, as the quotient space of $X \times Y \times [0,1]$ by the following equivalence relation

$$\forall x, x' \in X, \forall y, y' \in Y, (x,y,0) \sim (x,y',0) \text{ and } (x,y,1) \sim (x',y,1)$$

Intuitively, we join together the spaces X and Y with all possible segments between the two. Some examples of join include $X * \{y\}$ wich is the cone of X, $X * \mathbb{S}^0$ which is the suspension of X. More generally the join of X with $\mathbb{S}^n$ is an $n+1$-fold suspension.

By using the same ideas as in lemma 3, one can prove the following useful lemma.

Lemma 4. *Let $h : \mathbb{R}^k \times \mathbb{R}^{m-k} \to \mathbb{R}$ be a quasi homogeneous function such that for all $(y,z) \in \mathbb{R}^m$, we have $h(y,z) = a(y) + b(z)$ where a and b are quasi-homogeneous locally Lipschitz with respective exponents $\alpha, \beta > 0$. We have the following homotopy equivalence*

$$h^{-1}(-1) \simeq a^{-1}(-1) * b^{-1}(-1)$$

If we write $f(y,z) = \lambda(y) + q(z)$ as described by the normal form theorem, what remains to be done to understand the relative homology of (M_1, M_{-1}) is to describe $q^{-1}(-1)$ and $\lambda^{-1}(-1)$. The first one has the homotopy type of $\mathbb{S}^\mu$ where μ is the smooth Morse index of f, by classical Morse theory.

For the second, note that we have the following homotopy equivalence $\lambda^{-1}(-1) \simeq \lambda^{-1}(]-\infty, 0[)$. Using the min max form, we get

$$\lambda^{-1}(]-\infty,0[) = \bigcup_i \bigcap_{j \in S_i} \lambda_j(]-\infty,0[)$$

Note that each intersection is a convex cone we'll denote by C_i. We'll relate this to the simplical complex $K(S_1^c, ..., S_l^c)$ by using the nerve of the family $C = (C_i)_i$, i.e we'll look at the following set

$$N(C) := \{J \subset [\![1,l]\!] \mid \bigcap_{j \in J} C_j \neq \varnothing\}$$

The following lemma gives us that the nerve of $N(C)$ is exactly the one of the family of sets $S_1^c, ..., S_l^c$.

Lemma 5. *Let $J \subset [\![1,k+1]\!]$. J is the whole set if and only if*

$$\bigcap_{j \in J} \lambda_j^{-1}(]-\infty,0[) = \varnothing$$

Proof. We first show that if $J = [\![1,k+1]\!]$, we indeed get an empty intersection. Recall that 0 is a critical point of f, so it is a critical point of λ. We then have

$$0 \in \partial_0 \lambda \subset \mathrm{conv}\{\lambda_1, ..., \lambda_{k+1}\}$$

So there exists $\mu_1, ..., \mu_{k+1}$ with sum 1 such that $0 = \sum_i \mu_i \lambda_i$. We then see that all λ_i cannot be strictly negative at the same time.

We now show that if $J \neq [\![1,k+1]\!]$, the intersection is not empty. Condition (C3) gives us that the family $(\lambda_j)_{j \in J}$ is linearly independant. The linear system $\lambda_j(y) = -1, j \in J$ then has a solution, where the unknown is y. □

Because every C_i is convex, their intersections are contractible. Applying Leray's nerve theorem (cite), $\lambda^{-1}(-1)$ has the homotopy type of the simplical complex given by the nerve. By virtue of what we discussed, this complex is exactly $K(S_1^c, ..., S_l^c)$. The homotopy type of $f^{-1}(-1)$ is then given by

$$f^{-1}(-1) \simeq K(S_1^c, ..., S_l^c) * \mathbb{S}^\mu$$

We deduce the following result regarding homology

Theorem 5. *For $i \geq 0$, we have*

$$H_{i+\mu}(M_1, M_{-1}) = \tilde{H}_{i-1}(K(S_1^c, ..., S_k^c))$$

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